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Inventor(s)	Baker; Andrew R. et al.

Robot competition table

Abstract

A robot competition field assembly including field end sections, a field mid-section, and a field alignment system. The field end sections and the field mid-section include panels and sidewalls. The field alignment system is configured to align the panels so that there is a smooth transition between each of the playing surfaces of the field end sections and field mid-section.

Inventors: Baker; Andrew R. (Kokomo, IN), Blau; Daniel M. (Kokomo, IN), Kentfield; Jonathan J. (Kokomo, IN), Smith; Elisabeth C. (Kokomo, IN), Kinzie; Jay H. (Arlington Heights, IL)

Applicant: AndyMark, Inc. (Kokomo, IN)

Family ID: 81588119

Assignee: AndyMark, Inc. (Kokomo, IN)

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Primary Examiner: Ing; Matthew W

Attorney, Agent or Firm: Barnes & Thornburg

Background/Summary

PRIORITY CLAIM (1) This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application Ser. No. 63/115,332, filed Nov. 18, 2020, which is expressly incorporated by reference herein.

BACKGROUND

(1) Robotic competitions are events where robots compete against other robots in specific tasks to show various abilities and functions. Different competition games are set out on a competition field and competing robots move around the field to complete the game.

(2) Competition fields are often built to a regulated size on the site of a competition. These competition fields can be bulky and hard to store or transport. For example, robotic competitions held by FIRST LEGO League incorporate a regulated competition field that has a cumbersome size when in use. Accordingly, there is a need for competition fields that can be used for robotic competitions while allowing for easy storage and transport.

SUMMARY

(3) A field for robotic competitions according to this disclosure includes a number of sections assembled together for use so as to allow for disassembled storage and transportation. More specifically, an exemplary competition field includes field end-sections and at least one field mid-section. The competition also includes a field alignment system that provides means for locating

and coupling sections of the competition field so as to ensure smooth transitions along important surfaces and secure attachment of the sections when in use.

(4) In illustrative embodiments, sections of the competition field are made up of a panel that provides a playing surface and walls that surround the playing surface when assembled. The panels may be made from ply wood or other suitable material. The walls may be made from extruded plastic tube or other suitable material.

(5) In illustrative embodiments, the field alignment system includes brackets that extend inside walls between sections of the field and pins that extend inside panels between section of the field. The brackets function to align the walls and the playing surface of the competition field along outside edges of the competition field. The pins function to align panels that form the playing surface along the mid span of the competition field.

(6) In illustrative embodiments, a larger robotic competition field system may be provided by multiple robotic competition fields arranged side-by-side. Such systems can include coupling means for fixing robotic competition fields together in pre-set positions relative to one another. For example, a first robotic competition field and second robotic competition field may include integrated latches and catches that are engaged to fix the fields in position relative to one another.

(7) Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of illustrative embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

- (1) The detailed description particularly refers to the accompanying figures in which:
- (2) FIG. 1 is a perspective view of a robot competition field system that includes a robot competition field supported by a table so that the robot competition field is elevated off the ground, the view showing that the robot competition field includes three field sections that couple together via a field alignment system to form a smooth playing surface and a perimeter wall that extends around the smooth playing surface with smooth joints where the three field sections couple together;
- (3) FIG. 2 is an exploded perspective view of the robot competition field system of FIG. 1, showing the robot competition field has two field end sections and a field mid-section located between the field end sections, and further showing that side walls of the end sections contain brackets that extend into corresponding side walls of the field mid-section to align and couple the three field sections when assembled;
- (4) FIG. 3 is a detailed view of a portion of FIG. 2 showing the side walls of the field mid-section are rectangular extruded tubes sized to receive the brackets of the field end sections, and further showing that coupling edges of the field mid-section include a plurality of pin receiving apertures that receive alignment pins from the coupling edges of the field end sections;
- (5) FIG. 4 is a detailed view of a portion of FIG. 2 showing that the side walls of the field end sections are rectangular extruded tubes and the brackets are L-shaped with a vertical wall that extends along an inside edge of the side wall closest to the playing field and a horizontal wall that extends along a bottom surface of the side wall to couple the bracket to the side wall and end panel together, and further showing that coupling edges of the field end sections have a plurality of alignment pins that extend into the plurality of pin receiving apertures of the field mid-section to align the end panels and the mid-panel so that the robot competition field has a smooth surface across the joints of the field sections;
- (6) FIG. 5 is an exploded perspective view of the robot competition field of FIG. 2 showing that each of the end walls and side walls couple to perimeter edges of the end panels of the field end

sections and the brackets assemble into the ends of the side walls and couple therewith, and the side walls couple to the mid-panel of the field mid-section;

(7) FIG. 6 is a detailed perspective view of the robot competition field of FIG. 2 showing the bracket is assembled inside side walls of the field mid-section and the field end sections, the bracket is coupled with two bolts to the field end section and the bracket is coupled with an easy-assembly bolt to the field mid-section. FIG. 6 further shows that the sidewalls are coupled to the end panels and mid-panel using a bolt and flared nut that is located inside the rectangular extruded tube;

(8) FIG. 7 is a detailed view of the underside of the robot competition field of FIG. 6 showing the easy-assembly bolt uses a phillips screw head and the remaining bolts are located in recesses on the underside of the end panels and mid-panel;

(9) FIG. 8 is a top perspective view of the robot competition field of FIG. 1 showing one of the field end section coupled to the field mid-section at the corresponding coupling edges, and the other field end section spaced apart from the field mid-section so that the brackets and alignment pins of the field end section are aligned with the corresponding features of the field mid-section, and the field end section can be slid into position to couple with the field mid-section;

(10) FIG. 9 is a detailed perspective view of the robot competition field of FIG. 8 showing the bracket of the field end section is aligned with the rectangular aperture of the side wall of the field mid-section so that it can slide into the assembled position to couple therewith, and the alignment pins of the field end section are aligned with the pin receiving apertures of the field mid-section so that the field end section can slide into the assembled position to couple therewith;

(11) FIG. 10 is a detailed perspective view of two adjacent robot competition fields, and the first robot competition field includes an integrated latch at a terminal end of the end wall of the one of the field end sections, and the second robot competition field includes an integrated catch at a terminal end of the end wall of one of the field end sections so that the two robot competition fields can be coupled together, and

(12) FIG. 11 is a perspective view of two adjacent robot competition fields forming a robot competition field system on which one or two simultaneous games may be played with the integrated latches/catches of FIG. 10 fixing the fields together in a pre-set alignment relative to one another.

DETAILED DESCRIPTION

(13) An illustrative robot competition field assembly 10 includes a table 12 and a robot competition field 14 as shown in FIG. 1. The robot competition field 14 may be disassembled into three field sections 22, 24, 26 and has a field alignment system 16 that couples each of the field sections 22, 24, 26 together. The field alignment system 16 is adapted to provide a flat playing surface on which robot competition games can be played. Each of the three field sections 22, 24, 26 include panels 30, 52 that provide a portion of a playing surface, and endwalls 32 and sidewalls 34, 36, 54, 56 that form a perimeter wall around the flat playing surface. The field alignment system includes a plurality of brackets 70 that extend and couple between adjacent sidewalls 34, 36, 54, 56, and a plurality of alignment pins 72 that extend between adjacent coupling edges 48, 68 of each of the field sections 22, 24, 26 as shown in FIG. 2.

(14) The plurality of brackets 70 of the field alignment system 16 are L-shaped and extend into the sidewalls 34, 36 of the field end sections 22, 26 and are fixed to move therewith as shown in FIG. 6. Each of the brackets extend out of the sidewalls 34, 36 and into the sidewalls 54, 56 of the field mid-section 24 and couple with field mid-section 24 with an easy-assembly bolt 82. The plurality of brackets 70 and the plurality of alignment pins 72 provide strength and rigidity to the robot competition field 14 and align the edges of the panels 30, 52 and the sidewalls, 34, 36, 54, 56 so that the robot competition field 14 does not have any steps at the joints between the field sections 22, 24, 26.

(15) The robot competition field 14 is supported in an elevated position by the table 12 as shown in

FIG. 1. The table **12** is a folding table that includes a table top **18** and a plurality of legs **20**. The plurality of legs **20** extend downward from the table top **18** and engage the ground to keep the table top **18** spaced apart and elevated away from the ground. The robot competition field **14** is located on the table top **18**. In some embodiments, the table **12** may be replaced by a fixed table, a box, an eight-foot banquet table, a tote container, or other suitable support. In other embodiments, the robot competition field **14** may be placed directly on the ground.

(16) The robot competition field **14** provides a flat playing surface for robot competition games to be played on and has rectangular shape with length along a longitudinal axis **11** and width along a lateral axis **13** as shown in FIG. 1. The robot competition field **14** includes a first field end section **22**, a field mid-section **24**, and a second field end section **26** that couple together in series along the longitudinal axis **11** as shown in FIG. 2. This allows for the robot competition field **14** to be decoupled into smaller sections so that the robot competition field **14** is easier to transport and carry. Each of the field sections **22**, **24**, **26** may be sized so that when they are coupled together they form a regulation size competition field, but when they are decoupled they will fit inside a small vehicle.

(17) The first field end section **22** and the second field end section **26** are substantially similar and contain the same features as shown in FIG. 2. The first field end section **22** is located on one end of the robot competition field **14**, and the second field end section **26** is located on the opposite end of the robot competition field **14**. The field mid-section **24** is located between the first field end section **22** and the second field end section **26**. In some embodiments, the robot competition field **14** may include more than one field mid-section **24** so that each of the field sections have a smaller length along the longitudinal axis **11** and each field section may be more compact in the decoupled configuration for transport.

(18) The first and second field end sections **22**, **26** each include an end panel **30** and perimeter walls **28** that extend around three of the four edges of the end panel **30** as shown in FIGS. 2 and 3. The end panel **30** includes a playing surface **44**, a coupling edge **48**, and perimeter edges **46** that comprise the three edges of the end panel **30** that is not the coupling edge **48**. The playing surface **44** faces outward from the ground and is substantially flat. The coupling edge **48** extends along the lateral axis **13** and faces the field mid-section **24**.

(19) The perimeter walls **28** provide a barrier so that a robot will not fall off the robot competition field **14** when a game is taking place. The perimeter walls **28** include an endwall **32** and two sidewalls **34**, **36** as shown in FIGS. 2 and 3. The endwall **32** extends along the perimeter edge **46** opposite the coupling edge **48** of the end panel **30**. The endwall **32** extends outward from the playing surface **44** and is made from rectangular extruded tube. The endwall **32** is coupled to the end panel **30** with a plurality of flared nuts **40** and fixed bolts **42**. The sidewalls **34**, **36** extend along the longitudinal axis on the remaining two perimeter edges **46**. The sidewalls **34**, **36** extend outward from the playing surface **44** and are made from rectangular extruded tube. Each of the sidewalls **34**, **36** are perpendicular to the endwall **32** and abut opposite terminal ends of the endwall **32**. Each of the sidewalls **34**, **36** are coupled to the end panel **30** with a plurality of flared nuts **40** and fixed bolts **42**. The sidewalls **34**, **36** form assembly apertures **38** at the terminal ends of the sidewalls **34**, **36** at the coupling edge **48**.

(20) The field mid-section **24** includes a mid-panel **52** and two sidewalls **54**, **56** as shown in FIGS. 2 and 3. The mid-panel **52** includes a playing surface **66**, two coupling edges **68** that extend in the lateral direction, and two side edges **69** that extend in the longitudinal direction. The playing surface **66** faces outward from the ground and is substantially flat. The coupling edge **68** are on opposite sides of the mid-panel **52** and face each of the coupling edges **48** on the first and second field end sections **22**, **26**. Each of the coupling edges **68** include a plurality of receiving pin apertures **50** that extend into the mid-panel **52** and engage the field alignment system **16**.

(21) The sidewalls **54**, **56** extend along the longitudinal axis on side edges **69**. The sidewalls **54**, **56** extend outward from the playing surface **66** and are made from rectangular extruded tube. Each of

the sidewalls **54**, **56** are coupled to the mid-panel **52** with a plurality of flared nuts **40** and fixed bolts **42**. The sidewall **54** forms an assembly aperture **58** at one terminal end of the sidewall **54** that faces the first field end section **22** and an assembly aperture **60** at the other terminal end of the sidewall **54** that faces the second field end section **26**. The sidewall **56** forms an assembly aperture **62** at one terminal end of the sidewall **56** that faces the first field end section **22** and an assembly aperture **64** at the other terminal end of the sidewall **56** that faces the second field end section **26**.

(22) The field alignment system **16** couples the first field end section **22** to the field mid-section **24** on one side of the field mid-section **24**, and the second field end section to the field mid-section **24** on the opposite side of the field mid-section **24** as shown in FIGS. **1** and **2**. The field alignment system **16** aligns the playing surfaces **44**, **66** of each of the field sections **22**, **24**, **26** so that they form a substantially flat playing surface without steps at the joints between the field sections **22**, **24**, **26**. In the illustrative embodiment shown in the FIG. **8**, the first field end section **22** is coupled with the field mid-section **24**, and the second field end section **26** is aligned and ready to be coupled with the field mid-section **24**.

(23) The field alignment system **16** includes a plurality of brackets **70** and a plurality of alignment pins **72** as shown in FIGS. **2-4**, **6** and **9**. The plurality of brackets **70** includes a bracket that couples to each of the assembly apertures **58**, **60**, **62**, **64** of the field mid-section **24**. One of the brackets **70** is shown in the illustrative embodiment in FIG. **6**. The bracket **70** extends into the assembly aperture **38** of the sidewall **36** of the first field end section **22** and is coupled to the sidewall **36** with two fixed bolts **78** and two fixed nuts **80**. The bracket **70** protrudes out of the sidewall **36** of the first field end section **22** and into assembly aperture **62** of the sidewall **56** of the field mid-section **24** and is coupled to the sidewall **56** with an easy-assembly bolt **82** and an assembly nut **84**. In the illustrative embodiment, the easy-assembly bolt has a Phillips head configuration so that only a Phillips head screw driver is needed to assembly the robot competition field **14**. In other embodiments, the easy-assembly bolt may have a different fixing arrangement such as a flat head configuration, a hex-type configuration, a Torx-type configuration, or other bolt head configuration.

(24) One bracket **70** extends into the assembly aperture **38** of the sidewall **34** of the first field end section **22** and into assembly aperture **58** of the sidewall **54** of the field mid-section **24**. One bracket **70** extends into the assembly aperture **38** of the sidewall **34** of the second field end section **26** and into assembly aperture **60** of the sidewall **54** of the field mid-section **24**. One bracket **70** extends into the assembly aperture **38** of the sidewall **36** of the second field end section **26** and into assembly aperture **64** of the sidewall **56** of the field mid-section **24**.

(25) The bracket **70** has L-shape and includes a vertical leg **74** and a horizontal leg **76** as shown in FIG. **6**. The vertical leg **74** extends outwardly and adjacent to an inside wall of the sidewalls **36**, **56** so that the sidewalls **36**, **56** are aligned along the surface of the sidewalls **36**, **56** that faces the playing area of the robot competition field **14**. The horizontal leg extends in the lateral direction and adjacent to a bottom wall of the sidewalls **36**, **56** so that the mid-panel **52** and the end panel **30** are vertically aligned and the playing surface **44**, **66** have a smooth transition across the joint.

(26) The plurality of alignment pins **72** are equally spaced apart along the width of the coupling edges **48** of the first and second field end sections **22**, **26** as shown in FIGS. **2**, **8**, and **9**. The plurality of alignment pins **72** extend outwardly from the coupling edge **48** parallel to the longitudinal axis **11**. The plurality of alignment pins **72** are aligned with the plurality of receiving pin apertures **50** in the coupling edges **68** of the field mid-panel **52** as shown in FIG. **8**. When the field sections **22**, **24**, **26** are coupled together, the plurality of alignment pins **72** extend into the corresponding plurality of receiving pin apertures **50**.

(27) The L-shaped brackets **70** and the rectangular extruded tube sidewalls **34**, **36**, **54**, **56** provide additional rigidity to the mid-panel **52** and end panels **30** to prevent warping and bending along the length of the robot competition field **14**. The alignment pins **72** provide rigidity along the coupling edges **48**, **68** of the mid-panel **52** and the end panels **30** to prevent warping and bending along the width of the robot competition field **14**.

(28) Recesses **86** are provided on the underside of the mid-panel **52** and end panels **30** as shown in FIG. 7. The recesses **86** allow the fixed bolts **42**, **78** to be located within the recesses **86** so that the fixed bolts **42**, **78** do not protrude beyond the underside of the mid-panel **52** and the end panels **30** and the underside of the robot competition field **14** remains substantially flat. The easy-assembly bolts **82** extend through apertures in the mid-panel **52** and have countersinks so that the underside of the mid-panel **52** remains substantially flat.

(29) An integrated latch **88** is provided in one of the terminal ends of the endwall **32** and an integrated catch **90** is provided on the opposite terminal end of the endwall **32** as shown in FIG. 10. The integrated latch **88** may be used to couple to an integrated catch **90** in a second robot competition field that is adjacent to the first robot competition field **14** as shown in FIG. 11 to provide a robot competition field system **100**. Multiple robot competition field may be coupled together with the integrated latches **88** and integrated catches **90** in series in the lateral axis **13** direction. Integrated latches **88** may also hold two tables together in series in the longitudinal axis **11** direction.

(30) The present disclosure provides multiple advantages over the simple plywood competition tables currently used. The robot competition field **14** may be quickly disassembled into three field sections **22**, **24**, **26** for easy storage at home, in a classroom, or between tournaments. The field sections **22**, **24**, **26** may fit into small vehicles for transportation, or be carried by one adult or two students (each section may weigh about 22 lb). The field alignment system **16** couples the field sections **22**, **24**, **26** together to allow the playing surfaces to provide a consistent playing surface. The easy-assembly bolts allow for one Phillips Head Screwdriver to reassemble field sections **22**, **24**, **26** for use. The perimeter walls **28** and sidewalls **54**, **56** may be adjusted using the fixed bolts **42** and flared nuts **40** to allow for corrections to be made to their location relative to the mid-panel **52** and end panels **30**.

(31) The sidewalls **34**, **36**, **54**, **56** and endwalls **32** may be made from extruded black ABS plastic. The mid-panel **52** and the end panels **30** may be made from light weight MDF. The fixed bolts **42**, **78** may be rivnuts.

Claims

1. A robot competition field assembly, the assembly comprising a field elevator including a table top that has horizontal orientation and a plurality of legs coupled to table top and extending downwardly from the table top to keep the table top spaced apart from the ground, a competition field supported by the table top of the field elevator in an elevated position, the competition field including (i) a first field end section, (ii) a second field end section, and (iii) a field mid-section located between the first and second field end sections, the first and second field end sections each include an end panel having a playing surface, a coupling edge, and three perimeter edges, and three perimeter walls that each extend upwardly from the playing surface at each of the three perimeter edges, and the field mid-section includes a mid-panel having a playing surface, two coupling edges, and two side edges, and two side walls that extend upwardly from the playing surface at the two side edges, and a field alignment system that couples the first field end section to the field mid-section at the adjacent corresponding coupling edges and couples the second field end section to the field mid-section at the adjacent corresponding coupling edges, the field alignment system includes a plurality of brackets that extend from inside the perimeter walls at the coupling edge of the first and second field end sections to inside the side walls of the field mid-section and couple therewith, and a plurality of alignment pins that extend from inside the coupling edge of the first and second field end sections to inside the coupling edges of the field mid-section, wherein the field alignment system is configured to align the playing surfaces of the first and second field end sections and the field mid-section upon assembly of the robot competition field assembly so that a continuous playing field is formed without steps, and adjacent perimeter walls of the first and

second field end sections and side walls of the field mid-section form a smooth continuous wall along an assembled edge of the competition field without steps.

2. The assembly of claim 1, wherein the perimeter walls of the first and second field end sections include an end wall and a pair of side walls, and the end wall and the pair of side walls are made from rectangular extruded tube, and the two side walls of the field mid-section are made from rectangular extruded tube.

3. The assembly of claim 2, wherein the brackets are L-shaped and have a vertical leg that extends along an inside side face of each the side walls of the first and second field end sections and the field mid-section so that the side walls are aligned along the assembled edge of the competition field, and a horizontal leg that extends along a bottom inside surface of the sidewalls so that the playing surfaces of each of the first and second field end sections and the field mid-section are horizontally aligned.

4. The assembly of claim 3, wherein the alignment pins include a plurality of pins spaced along a width of the coupling edges of the end panels, and the mid-panel includes a plurality of pin receiving apertures along the coupling edges that are aligned with the plurality of pins of the end panels.

5. The assembly of claim 3, wherein the brackets are fixed inside the side walls of the first and second field end sections when the first and second field end sections are decoupled from the field mid-section, and assembly bolts couple the brackets to each corner of the field mid-section when the first and second field end sections are coupled to the field mid-section.

6. A robot competition field comprising field end sections each including an end panel made from wood and having a playing surface, a pair of side edges, an end edge, and a coupling edge, an end wall that extends upward from the end edge of the end panel, and sidewalls that extend upward from the side edges of the end panels, wherein the end walls and the sidewalls that extend upward from the side edges of the end panels are made from rectangular extruded tube, a field mid-section including a mid-panel made from wood and having a playing surface, coupling edges, and side edges, and sidewalls that extend upward from the side edges of mid-panel, wherein the sidewalls that extend upward from the side edges of mid-panel are made from rectangular extruded tube, and a field alignment system configured to couple the coupling edges of the field end sections to the field mid-section upon assembly of the robot competition field, the field alignment system including brackets that extend from inside the sidewalls of the field end sections to inside the sidewalls of the field mid-section to couple the field end sections and the field mid-section together, and alignment pins that extend from inside the end panels of the field end sections to inside the mid-panel of the field mid-section at each of the corresponding the coupling edges, wherein the coupling edges are configured to be engaged and aligned so that there is a smooth transition between each of the playing surfaces upon assembly of the robot competition field, wherein the brackets are L-shaped and have a vertical leg that extends along an inside side face of each the side walls of the field end sections and the field mid-section so that the sidewalls are aligned along the assembled edge of the competition field, and a horizontal leg that extends along a bottom inside surface of the sidewalls so that the playing surfaces of each of the field end sections and the field mid-section are horizontally aligned.

7. The robot competition field of claim 6, wherein the alignment pins include a plurality of pins spaced along a width of the coupling edges of the end panels, and the mid-panel includes a plurality of pin receiving apertures along the coupling edges that are aligned with the plurality of pins of the end panels.

8. The robot competition field of claim 6, wherein the brackets are fixed inside the sidewalls of the field end sections when the field end sections are decoupled from the field mid-section, and assembly bolts couple the brackets to each corner of the field mid-section when the field end sections are coupled to the field mid-section.

9. A robot field competition system, the system comprising a first robot competition field including

(1) field end sections each including an end panel made from wood and having a playing surface, a pair of side edges, an end edge, and a coupling edge, an end wall that extends upward from the end edge of the end panel, and sidewalls that extend upward from the side edges of the end panels, wherein the end walls and the sidewalls are made from rectangular extruded tube, (2) a field mid-section including a mid-panel made from wood and having a playing surface, coupling edges, and side edges, and sidewalls that extend upward from the side edges of mid-panel, wherein sidewalls are made from rectangular extruded tube, and (3) a field alignment system that couples the coupling edges of the field end sections to the field mid-section and includes brackets that extend from inside the sidewalls of the field end sections to inside the sidewalls of the field mid-section to couple the field end sections and the field mid-section together, and alignment pins that extend from inside the end panels of the field end sections to inside the mid-panel of the field mid-section at each of the corresponding the coupling edges, wherein the coupling edges are configured to be engaged and aligned so that there is a smooth transition between each of the playing surfaces upon assembly of the robot competition field, a second robot competition field including (1) field end sections each including an end panel made from wood and having a playing surface, a pair of side edges, an end edge, and a coupling edge, an end wall that extends upward from the end edge of the end panel, and sidewalls that extend upward from the side edges of the end panels, wherein the end walls and the sidewalls that extend upward from the side edges of the end panels are made from rectangular extruded tube, (2) a field mid-section including a mid-panel made from wood and having a playing surface, coupling edges, and side edges, and sidewalls that extend upward from the side edges of mid-panel, wherein the sidewalls that extend upward from the side edges of mid-panel are made from rectangular extruded tube, and (3) a field alignment system that couples the coupling edges of the field end sections to the field mid-section and includes brackets that extend from inside the sidewalls of the field end sections to inside the sidewalls of the field mid-section to couple the field end sections and the field mid-section together, and alignment pins that extend from inside the end panels of the field end sections to inside the mid-panel of the field mid-section at each of the corresponding the coupling edges, wherein the coupling edges are engaged and aligned so that there is a smooth transition between each of the playing surfaces, wherein an integrated latch is provided in a first terminal end of the end wall of the first robot competition field and an integrated catch is provided on a second terminal end of the end wall of the second robot competition field, and wherein the integrated latch is configured to be engaged with the integrated catch to fix the first robot competition field to the second robot competition field upon assembly of the robot field competition system.

10. The system of claim 9, wherein the brackets are L-shaped and have a vertical leg that extends along an inside side face of each the sidewalls of the field end sections and the field mid-section so that the sidewalls are aligned along the assembled edge of the competition field, and a horizontal leg that extends along a bottom inside surface of the sidewalls so that the playing surfaces of each of the field end sections and the field mid-section are horizontally aligned.

11. The system of claim 10, wherein the alignment pins include a plurality of pins spaced along a width of the coupling edges of the end panels, and the mid-panel includes a plurality of pin receiving apertures along the coupling edges that are aligned with the plurality of pins of the end panels.

12. The system of claim 10, wherein the brackets are fixed inside the sidewalls of the field end sections when the field end sections are decoupled from the field mid-section, and assembly bolts couple the brackets to each corner of the field mid-section when the field end sections are coupled to the field mid-section.

13. The system of claim 9, wherein the sidewalls along one side of the first robot competition field contact the sidewalls along one side of the second robot competition field while the integrated latch is engaged with the integrated catch.

14. The system of claim 13, wherein the playing surfaces of the first robot competition field are co-

planar with the playing surfaces of the second robot competition field while the integrated latch is engaged with the integrated catch.
